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Optical Manipulation of Nanoparticles: A Selective Excitation Approach Using Highly Focused Orbital Angular Momentum Beams

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Orbital angular momentum and polarization states of highly focused vector vortex beams can be engineered to selectively excite the desired multipoles in nanoparticles, making them ideal candidates for complex optical applications. A platform based on the generalized Lorenz–Mie theory integrated with the complex source point method is developed to model the interaction of such beams with particles analytically. The existing platform is extended for obtaining the full-vector electromagnetic fields, outlining the observed effects that adding a substrate brings to the problem space. Despite the cross-coupling between different multipoles induced by the substrate, it is concluded that the proper choice of beam parameters enables the selective excitation of multipoles even in the critical case of a plasmonic substrate. The proposed formalism is general and allows a multipole study of the optical forces. Selective excitation is exploited to control the imparted optical forces on particles placed on a plasmonic substrate. 3D trapping regions are achievable for highly absorptive and high-refractive-index particles. Motion trajectory analyses are performed to demonstrate the trapping stability, concluding that highly focused optical beams, owing to the exceptional selective excitation ability, are suitable candidates for novel optical manipulation applications.

1. Introduction

Over the past decades, highly focused optical beams (HFOBs) have been extensively explored by the nanooptics scientific community to achieve high-intensity fields in small regions of space. The strong electric field generated by the HFOB at the focal plane is highly desirable for optical trapping,[1–5] microscopy,[6] and particle sizing,[7] among other applications. To ensure a reliable functionality, choosing the appropriate polarization for the HFOB is essential since it directly affects the minimum feasible focusing size. The use of azimuthally and radially polarized vector beams, particular forms of spirally polarized beams, is appealing because of their capability to tight focus, achieving a remarkably smaller spot size than the linearly polarized beams.[8–10] Furthermore, spirally polarized vector beams can be expressed by a couple of points on the surface of the Poincare sphere, unlike the conventional polarization cases where a single point can represent them.[11] Generally, HFOBs can be generated using precisely chosen high-numerical-aperture (NA) bulk lenses and metasurfaces to focus the incident beam's wavefront; however, perceiving rigorous analytic modeling of such beams can be a challenging task.[11–13] Unlike optical systems with small numerical apertures where the scalar paraxial solutions are adequate to model the beam's behavior,[14] high-numerical-aperture systems require taking the polarization into account, owing to its significant effects on the size and shape of the beam intensity after focusing.[15] The basic formalism to analyze highly focused vector beams was initially introduced by Richards and Wolf[16] and has been extended later in different studies on tight focusing beams, both theoretically[17–19] and experimentally.[20–22]

The ability of carrying orbital angular momentum (OAM) is one of the prosperous aspects of optical fields, which can offer a variety of applications in nanophotonics instruments.[23–28] In addition, beams carrying OAM can be characterized using an integer or semiinteger constant called the topological charge (m), defining the phase singularity order at the beam center, translating to a spiral phase profile as illustrated in **Figure 1**a. Figure 1 also shows the amplitude (up) and phase (bottom) profiles for the azimuthally (b) and radially (c) polarized beams with $m = +1$.[29] One of the well-known approaches for generating an OAM beam is creating a phase singularity (also known as optical

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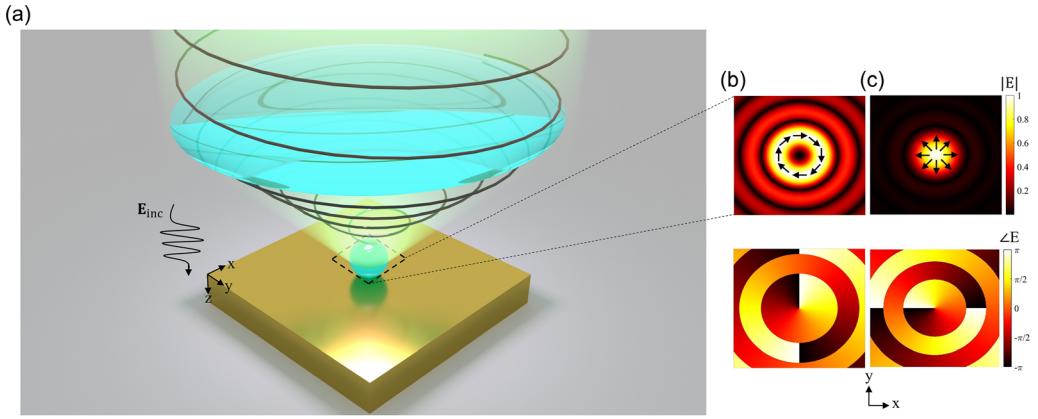


Figure 1. a) Schematic depiction of the HFOB carrying OAM acting on a nanoparticle above the gold substrate. The focal plane phase and amplitude profiles of the highly focused vector vortex beams associated with the topological charge of $m = +1$ for b) azimuthal and c) radial polarization.

vortex) in the light field.[30] At the singularity, the phase is undefined, and the beam intensity vanishes, resulting in a hollow or dark beam carrying OAM,[31] which is of interest in various applications.[32,33]

Recently, there has been an increasing interest in studying the interaction of highly focused OAM beams with metallic and dielectric nanoparticles due to the possibility of selectively exciting the desired multipoles through engineering the vortex beam profile, such as polarization and topological charge.[34–39] To be more specific, it has been demonstrated that the electric or magnetic multipoles of a silicon nanosphere can be selectively excited by choosing the radial or azimuthal polarization, respectively.[36,39] Besides, based on the total angular momentum delivered to the object by optical vortices, it has been proven that high-order multipoles can be selectively excited,[37,38] which shows the potential of the beams carrying OAM for higher-order selective multipole excitation in nanoparticles. Momentum exchange of light with the matter can exert radiation pressure and optical gradient forces on the object, which has been exploited for different applications, including optical manipulation[40–43] and optical levitation of macroscopic objects and accelerating relativistic lightsails.[44–51] Besides, HFOBs with different polarization and OAM states have been used for diverse nanoparticle manipulation applications, such as optical trapping[52] and tweezers.[53] The complex beam profile of HFOBs can provide optical gradient forces that compete with the radiation pressure, resulting in a compelling optical force response.[54] Moreover, the selective multipole excitation enabled by the HFOBs carrying OAM introduces a new platform to understand and engineer the contribution of different multipoles to the induced optical forces.[55]

Nevertheless, researchers need to rely on numerical simulations[56,57] to explore the behavior of HFOBs containing OAM due to the complexity of analytical solution and modeling of the electric fields since only the complete vector solution can precisely represent an HFOB. Describing the beams in terms of electromagnetic multipoles is one of the approaches, which can be used to overcome the mentioned issue. Decomposing the incident field in terms of multipoles is particularly valuable for studying the interaction of HFOBs carrying OAM with nanoobjects, as it enables utilization of the generalized Lorenz–Mie theory (GLMT) to obtain the desired electromagnetic fields.[58,59] The GLMT provides a linear mapping between an arbitrary incident field's beam shape coefficients (BSCs) and the scattered field coefficients of a spherical particle. However, it is not restricted to the spherical particles; it has been generalized for arbitrary geometries using the T-matrix method.[60,61] Another valuable advantage of expanding the fields in terms of multipoles is rigorous and efficient calculation of Maxwell stress tensor and optical forces.[55] Moreover, the multipole expansion of the fields yields valuable physical insight into the light–particle interaction.[62–65] It is also a beneficial analytical platform to compute the scattering characteristics of various ranges of particles.[66–69]

The first step in studying the interaction of a nanoparticle with a vortex HFOB through the GLMT platform is to find the BSCs of the incident field. Several numerical techniques have been proposed to determine the multipole expansion of highly focused vortex beams.[17,70–72] In, ref.[17] the BSCs are derived for a general case of nonuniformly polarized highly focused vector beams using the angular spectrum representation of the fields, demonstrating an excellent agreement between the theory and measurements. Moreover, another approach could be matching the expanded field to the exact one on a particular surface[70,71] or at the far field.[72] It should be noted that all the mentioned methods better reproduce the electric field distribution of a strongly focused laser beam in the stated regions. Despite possessing a satisfactory performance, numerical calculation of the BSCs enhances the computational burdens. Therefore, an analytical approach to express the BSCs of highly focused vector vortex beams in the focal region has been proposed in other studies,[73–76] where the authors have resorted to the complex source point (CSP) method to analytically describe the vortex HFOB for different polarization near the focusing area.

In this work, we propose utilizing the CSP method to generate a set of solutions satisfying the exact vector Helmholtz equation, which can be expanded analytically to express the BSCs of an HFOB containing OAM. Moreover, we extend the present analytical formalism to precisely calculate the full-vector electric field in the presence of a planar substrate. The formalism allows for

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analytical calculation of the scattering characteristics and induced optical forces of a spherical nanoparticle placed close to a substrate, as shown schematically in Figure 1. Our proposed procedure is not only more physically insightful but also computationally efficient and can be applied to nanoparticles with various sizes and materials over layered substrates. The numerical accuracy of the proposed model is verified by performing the convergence test over the number of multipoles contributing to the solution. Besides, the results are verified using the Lumerical FDTD package, and it has been demonstrated that the results of our analytical model are in good agreement with those of the FDTD (see Section S12, Supporting Information). Having provided the formalism, we apply it to study the interaction of the HFOBs with OAM with different nanoparticles. Our results show that the field profile of the HFOBs with various polarization and OAM states can be engineered to selectively excite the electrical or magnetic multipoles of the particle. Specifically, we illustrate that using the appropriate beam profiles, it is possible to selectively excite the multipoles in a silicon nanosphere that supports three different multipoles in the visible light region. We show that even in scenarios where the multipole resonances occur at the same wavelength (for instance, in a coated silicon sphere), the order and polarization of the incident beam can be properly chosen to selectively excite the desired multipole, which cannot be accomplished using a plane or Gaussian wave profile. It is also shown that selective excitation maintains its performance in the presence of a planar gold substrate. After exploring the characteristics of each multipole after excitation, we use our approach to calculate the optical forces acting on the gold and silicon nanospheres above a gold substrate, as examples of highly absorptive and high-refractive-index objects. It is illustrated that both particles can be trapped in three dimensions over the substrate. Also, we investigate the motion trajectory of the particles in the scenarios, in which the particle can be trapped in three dimensions to show the trapping stability. We demonstrate that a stable 3D trapping can be obtained using the azimuthally polarized HFOBs carrying OAM. These findings could pave the way for new photonic devices for selective excitation and trapping/tweezers.

The rest of this article is organized as follows. Section 2 briefly describes the theoretical formalism used to generate the results. In Section 3, the scattering properties of the proposed nanoparticles are investigated under a plane wave illumination; besides, the selective excitation using highly focused vortices is studied for the cases of silicon and multilayered nanoparticles. In Section 4, we investigate the optical forces resulting from the interaction of HFOB with different polarization and OAM states with particles placed on the gold substrate; we also study the trapping possibility in three dimensions. Section 5 gives the conclusion.

2. Analytical Formalism

This section briefly describes the theoretical formalism employed in this manuscript to calculate the electromagnetic and optomechanical results for nanoparticles illuminated by an HFOB with OAM, as shown in Figure 1. We proceed to derive the scattering characteristics and optical forces. Finally, we finish this section by calculating the BSCs for the incident beams used in the paper.

2.1. Scattering from a Nanoparticle over a Planar Substrate

In the proposed setup (Figure 1), the total field ($\mathbf{E}_{\mathrm{T}}$) outside the particle can be written, as

$$\mathbf{E}_{\mathrm{T}} = \mathbf{E}_{\mathrm{inc}} + \mathbf{E}_{\mathrm{inc,r}} + \mathbf{E}_{\mathrm{sc}} + \mathbf{E}_{\mathrm{sc,r}} \tag{1}$$

where $\mathbf{E}_{\mathrm{inc}}$ and $\mathbf{E}_{\mathrm{inc,r}}$ are the primary and secondary (reflected) incident fields, respectively, and $\mathbf{E}_{\mathrm{sc}}$ and $\mathbf{E}_{\mathrm{sc,r}}$ are the primary and secondary scattered fields from the nanoparticle. Note that the secondary fields are a result of the presence of the substrate. Next, to compute the total field using the GLMT, all the field contributions need to be expanded in terms of the electromagnetic multipoles as follows.

$$\mathbf{E}_{\iota} = \sum_{l=1}^{\infty}\sum_{m=-l}^{+l}\left[a_{\iota l}^{\ m}\mathbf{M}_{l}^{\mathrm{r}m} + b_{\iota l}^{\ m}\mathbf{N}_{l}^{\mathrm{r}m}\right], \qquad \iota = \{\mathrm{inc}-\mathrm{inc}, r-\mathrm{sc}, r\} \tag{2}$$

$$\mathbf{E}_{\mathrm{sc}} = \sum_{l=1}^{\infty}\sum_{m=-l}^{+l}\left[a_{\mathrm{sc}l}^{\ \mathrm{m}}\mathbf{M}_{l}^{\mathrm{s}m} + b_{\mathrm{sc}l}^{\ \mathrm{m}}\mathbf{N}_{l}^{\mathrm{s}m}\right] \tag{3}$$

where ι refers to $\mathbf{E}_{\mathrm{inc}}$ ($\iota = \mathrm{inc}$), $\mathbf{E}_{\mathrm{inc,r}}$ ($\iota = \mathrm{inc, r}$), or $E_{\mathrm{sc,r}}$ ($\iota = \mathrm{sc, r}$), m and l are integers, $\mathbf{M}^{\mathrm{r}}(\mathbf{M}^{\mathrm{s}})$ and $\mathbf{N}^{\mathrm{r}}(\mathbf{N}^{\mathrm{s}})$ are the regular (singular) electric and magnetic multipoles, respectively, and $a_{\iota l}^{\ \mathrm{m}}$ and $b_{\iota l}^{\ \mathrm{m}}$ are the multipole moments. The definition of the multipoles used in our article is given by another study[77] and also can be found in detail in the Supporting Information.

Having written all the field contributions in terms of the electromangnetic multipoles, the next step is associating the incident and scattered fields with their reflection from the planar substrate. Therefore, the fields need to be transformed to plane waves, opening up the possibility of using the well-known Fresnel coefficients to find the reflected fields. Finally, the reflected inhomogeneous plane waves are supposed to be converted back in terms of the multipoles. After some mathematical manipulations, the mapping from the primary to the secondary coefficients is given by the expressions below[78–80]

$$a_{\zeta,rl}^{\ \mathrm{m}} = \sum_{l'=l}^{\infty}\sum_{m'=-l'}^{+l'}\left[{}_{\zeta}^{aa}R_{ll'}^{mm'}a_{\zeta l'}^{\ m'} + {}_{\zeta}^{ab}R_{ll'}^{mm'}b_{\zeta l'}^{\ m'}\right] \tag{4}$$

$$b_{\zeta,rl}^{\ m} = \sum_{l'=l}^{\infty}\sum_{m'=-l'}^{+l'}\left[{}_{\zeta}^{ba}R_{ll'}^{mm'}a_{\zeta l'}^{\ m'} + {}_{\zeta}^{bb}R_{ll'}^{mm'}b_{\zeta l'}^{\ m'}\right] \tag{5}$$

where ζ refers to the primary incident ($\zeta = \mathrm{inc}$) or scattered ($\zeta = \mathrm{sc}$) fields. The mapping coefficients and their detailed derivation can be found in the Supporting Information.

To numerically calculate the unknown coefficients of the fields using the GMLT, it is required to truncate the summation over l to N_{T} terms, so all the coefficients ($a_{\zeta l}^{\ m}$ and $b_{\zeta l}^{\ m}$) will be vectors with finite $N_{\mathrm{T}} \times (2N_{\mathrm{T}}+1)$ dimension. Accordingly, by manipulating Equation (1), the primary and secondary scattered field coefficients can be written in terms of a linear equation as follows.

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$$\begin{bmatrix}\mathbf{a}_{sc}\\ \mathbf{b}_{sc}\end{bmatrix} = \mathbf{A}\left(\begin{bmatrix}\mathbf{a}_{inc}\\ \mathbf{b}_{inc}\end{bmatrix} + {}_{inc}\mathbf{R}\begin{bmatrix}\mathbf{a}_{inc}\\ \mathbf{b}_{inc}\end{bmatrix} + {}_{sc}\mathbf{R}\begin{bmatrix}\mathbf{a}_{sc}\\ \mathbf{b}_{sc}\end{bmatrix}\right) \tag{6}$$

in which $\mathbf{A}$ is a matrix containing the Mie coefficients (provided in Section S2, Supporting Information), and ${}_{\zeta}\mathbf{R}$ is the mapping matrix from the primary incident and scattered fields to the secondary ones, given in the following

$${}_{\zeta}\mathbf{R} = \begin{bmatrix}{}^{aa}_{\zeta}\mathbf{R} & {}^{ab}_{\zeta}\mathbf{R}\\ {}^{ba}_{\zeta}\mathbf{R} & {}^{bb}_{\zeta}\mathbf{R}\end{bmatrix} \tag{7}$$

with its entries given in the Supporting Information. In fact, the substrate brings out cross-coupling between multipoles with different order, which can be modelled mathematically using the matrix given in Equation (7).

Since the incident wave coefficients (BSCs) $a_{inc}{}^{m}$ and $b_{inc}{}^{m}$ are known, the scattered field coefficients are calculated by solving the linear system Equation (6).

After this procedure, the full electric field is known, and it is possible to calculate the scattering characteristics of the particles. The scattering time-averaged Poynting vector $\mathbf{S}_{sca}$ can be written as

$$\mathbf{S}_{sca} = \frac{1}{2}\Re\{(\mathbf{E}_{sc} + \mathbf{E}_{sc,r}) \times (\mathbf{H}_{sc} + \mathbf{H}_{sc,r})^{*}\} \tag{8}$$

and the total scattered power is computed by integrating $\mathbf{S}_{sca}$ over a closed surface enclosing the particle (A)

$$P_{sca} = \oiint_{A}\mathbf{S}_{sca}\cdot\mathbf{dr}^{2} \tag{9}$$

Finally, the scattering cross section C_{sca} can be computed as follows

$$C_{sca} = \frac{P_{sca}}{P_{0}} \tag{10}$$

in which P_0 is the power of the incident field. It is worth noting that the scattering results are normalized in our manuscript for the sake of easier comparison between different illuminations. To see the actual values for scattering cross sections, the reader is referred to Section S11, Supporting Information.

Moreover, to compute the optical forces acting on the particle, we use the Maxwell stress tensor (MST) method, which can be written, as follows[42,45,49]

$$\mathbf{T}_{M} = (\varepsilon_{0}\varepsilon_{r}\mathbf{E}\otimes\mathbf{E}^{*} + \mu_{0}\mathbf{H}\otimes\mathbf{H}^{*}) + \frac{1}{2}(\mathbf{E}\times\mathbf{E}^{*} + \mathbf{H}\times\mathbf{H}^{*}) \tag{11}$$

in which ε_r is the permittivity of the surrounding medium, and $\otimes$ is the tensor product between two vectors. After defining the proper MST, one may find all the time-averaged optical forces acting on the nanoparticle by integrating the MST over a closed surface enclosing the scatterer (A) as shown below

$$\langle\mathbf{F}\rangle = \frac{1}{2}\Re\left\{\oiint_{A}\mathbf{T}_{M}\cdot\mathbf{dr}^{2}\right\} \tag{12}$$

It is noteworthy that the closed form of the equations needed for the computation of the optical forces can be found in Section S9, Supporting Information.

2.2. Complex Source Point Method to Derive the BSCs

To find the BSCs for a plane wave used to explore the scattering characteristics of the nanoparticles in our paper, the reader is referred to read Section S3, Supporting Information. For the HFOBs carrying OAM, we use the CSP method to find the BSCs.[76] In the CSP method, the position vector is translated on the propagation axis to the complex plane, and the absolute value of the complex position vector is given by

$$r_{c} = \sqrt{x^{2} + y^{2} + (z - iz_{R})^{2}} \tag{13}$$

where $z_R = \frac{kw_0^2}{2}$ is the Rayleigh length of the beam, k is the free space wave number, and w_0 is the beam waist. The solution of the scalar Helmholtz equation in the complex plane $(\psi_{c\nu}^{\mu})$ expresses a scalar beam as given below.

$$\psi_{c\nu}^{\mu}(kr_{c}) = A_{0}j_{\nu}(kr_{c})P_{\nu}^{\mu}(\cos\theta_{c})e^{i\mu\phi} \tag{14}$$

where $\theta_c = \arctan\left(\frac{\sqrt{x^2+y^2}}{z-iz_R}\right)$, and by choosing $\nu = |\mu|$, the profile represents a scalar highly focused Laguerre–Gaussian scalar beam. Finally, the beam can be expanded in terms of the scalar multipoles using the translation coefficients[81] as

$$a_{l}^{m} = \delta_{m\mu}A_{0}(2l+1)(2\nu-1)!!\frac{j_{l}(ikz_{R})}{(ikz_{R})^{\nu}}, \quad m \geq 0 \tag{15}$$

$$a_{l}^{m} = \delta_{m\mu}A_{0}(2l+1)\frac{1}{2^{\nu}\nu!}\frac{(l+\nu)!\,j_{l}(ikz_{R})}{(l-\nu)!\,(ikz_{R})^{\nu}}, \quad m \leq 0 \tag{16}$$

where $\delta_{m\mu}$ is the Kronecker delta function, $(2\nu-1)!! = \frac{2^{-\nu}(2\nu)!}{\nu!}$, and $A_0 = \frac{(-ikz_R)^{\nu}}{j_{\nu}(-ikz_R)}$ is a normalization constant. The vector beams can be constructed using the scalar solutions, and the BSCs for the magnetic and electric vortex beams with the toplogical charge of $m = \mu$ are given, respectively, by

$$\mathbf{E}_{inc} = \mathbf{V}_{M\nu}^{\mu} = \sum_{l=\nu}^{\infty}\sum_{m=-l}^{+l}\frac{a_{l}^{m}}{K_{l}^{m}}\mathbf{M}_{l}^{rm} \tag{17}$$

$$\mathbf{E}_{inc} = \mathbf{V}_{E\nu}^{\mu} = \sum_{l=\nu}^{\infty}\sum_{m=-l}^{+l}\frac{a_{l}^{m}}{K_{l}^{m}}\mathbf{N}_{l}^{rm} \tag{18}$$

where K_l^m is the multipoles normalization constant given in Equation (S19) of the Supporting Information, and $\mathbf{V}_{M\nu}^{\mu}$ ($\mathbf{V}_{E\nu}^{\mu}$) is the magnetic (electric) beam with the topological charge of $m = \mu$. It is also worth noting that these beams represent the conventional azimuthally and radially polarized beams for $\nu = \mu = 0$.[75] For a complete treatment of the problem, see Supporting Information.

3. Selective Excitation of Multipoles

Here, we introduce an approach that describes the possibility of exciting a specific resonance mode of a nanoparticle by engineering the beam profile of the vortex HFOBs. However, in order to fully understand the selective excitation concept, it is crucial to

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determine the multipole modes supported by each nanoparticle. Therefore, we divide this section into two parts: first, we illuminate the particles with a simple plane wave that reveals the supported multipoles and their cross-coupling in the presence of the substrate; then, we use the HFOBs carrying OAM to excite a specific multipole.

3.1. Scattering Properties of the Nanoparticles under a Plane Wave Illumination

To provide the infrastructural knowledge of the interaction of light with the proposed particles, we exploit the analytical formalism introduced in the previous section to investigate the scattering characteristics of the nanoparticles in the absence and presence of a planar metallic substrate. We start by exploring the interaction of a plane wave with three particles consisting of gold and silicon. The first two nanoparticles are spheres purely made of gold and silicon with radius $a = 90$ nm to have all the supported multipoles in the visible range.[42] The third particle is a layered sphere optimized to have electric and magnetic resonances at the same wavelength.[82] It consists of a silicon core with a radius of 74 nm, a middle shell made of gold with a thickness of 11 nm, and an outer silicon cover with a width of 5 nm. The refractive index of gold is taken from Johnson and Christy data,[83] and the refractive index of silicon is assumed to be constant $n_{Si} = 3.4$. The plane wave propagates in the $z-$direction associated with an x-directed electric field.

The total normalized scattering cross sections (solid green line) and its decomposition into magnetic dipole (MD, blue circle), electric dipole (ED, red stars), magnetic quadrupole (MQ, purple circles), and electric quadrupole (EQ, sky blue stars) for the gold, silicon, and shelled nanoparticles without the substrate are shown in **Figure 2**a–c, respectively. As expected, the gold nanosphere only supports ED with a resonance at $\lambda = 588$ nm within the visible range.[84] The contribution from higher-order electric and magnetic multipoles in the scattering is negligible, as shown in Figure 2a. However, the silicon nanoparticle supports three distinctive multipoles in the visible range,[85] as shown in Figure 2b. The MD and ED resonances occur at $\lambda = 640$ nm and $\lambda = 492$ nm, respectively, while the MQ resonance is placed at $\lambda = 443$ nm. It should be noted that the quality factor of the MQ resonance is the highest compared to the ED and MD resonances. The quality factor is directly related to the power absorbed or radiated by the particle,[86] and the more the power couples to the far field, the lower the quality factor is. As illustrated in Figure 2b, the ED demonstrates the best

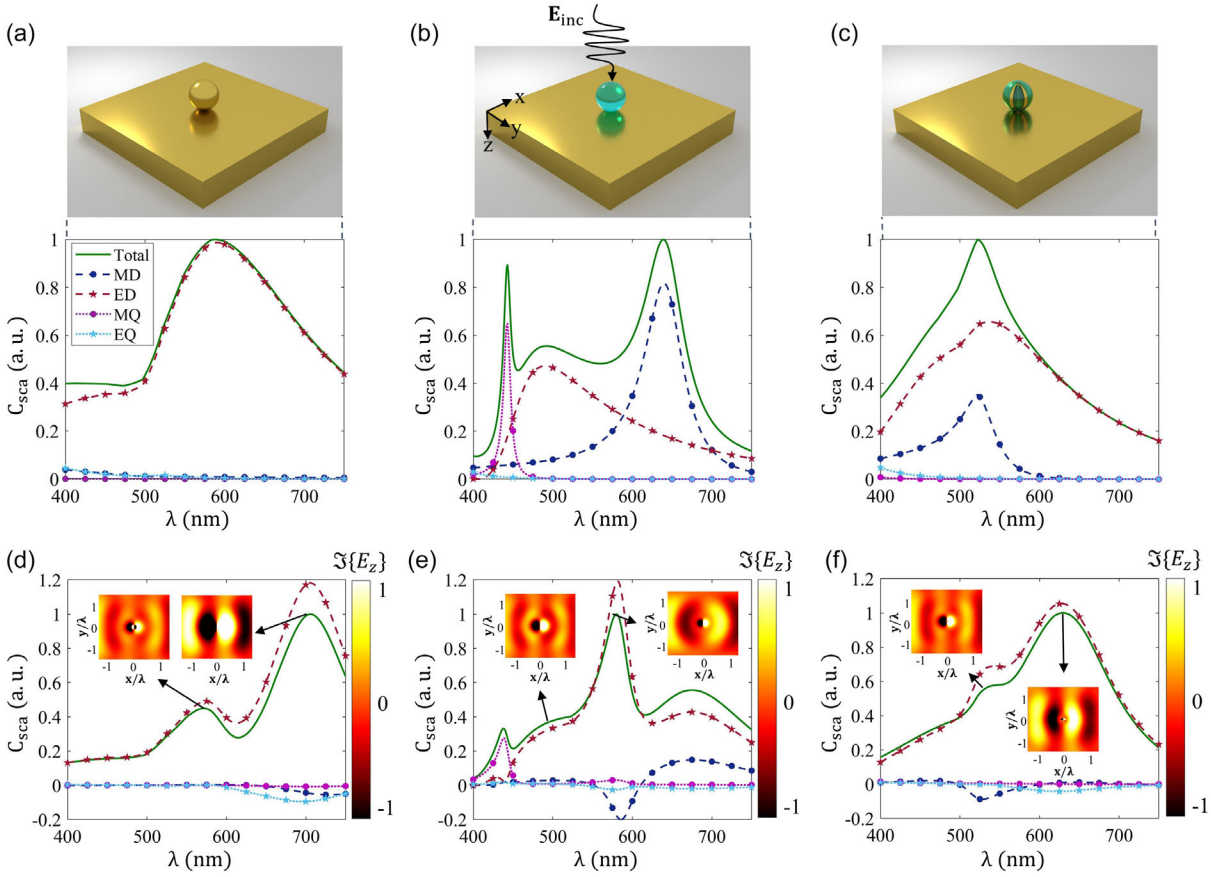


Figure 2. The normalized scattering cross section and its decomposition into the electromagnetic multipoles for the a,d) gold, b,e) silicon, and c,f) multilayer nanoparticles in a–c) free space and d–f) on top of a planar gold substrate under a plane wave illumination. It should be noted that the near fields within (d–f) are plotted on the substrate to determine the origin of different resonances in the ED contribution. The figure shows the scattering characteristics of the proposed nanoparticles in the visible light range.

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coupling to the far-field radiation for the silicon nanosphere, resulting in a lower quality factor.[87] The MD and MQ modes present poor coupling, resulting in higher quality factors compared to the ED mode. Furthermore, as the gold nanoparticle (Figure 2a) delivers a significant absorption in the visible range, the quality factor is highly reduced with respect to the ED resonance mode in the silicon nanoparticle. Finally, for the multilayer nanoparticle, the total normalized scattering cross section with its multipole decomposition is given in Figure 2c. As shown, both ED and MD coincide at the same wavelength ($\lambda = 522$ nm), while the quality factor of the MD is higher than the ED due to the same reasons explained for the previous nanoparticles.

Next, due to its importance in many experiments,[40,79,88] we investigate the impact of placing a metallic substrate under the nanoparticles. As explained in Section 2, the substrate introduces cross-coupling between different excited multipole modes. Moreover, the interface between the gold substrate and free space supports the propagation of surface plasmon polaritons (SPPs), which significantly affects the scattering behavior of the nanoparticles. The calculated normalized scattering cross sections and their multipole decompositions for all the nanoparticles are shown in Figure 2d–f. The gold substrate induces an additional resonance at the scattering spectra of the gold sphere at $\lambda = 706$ nm, as expressed in Figure 2d. Unlike the ED resonance at $\lambda = 572$ nm, the SPP resonance is mainly due to the strong coupling between the ED scattering and the propagating surface waves. The insets in Figure 2d show the z-component of the electric field on the surface of the substrate at $\lambda = 572$ nm and $\lambda = 706$ nm. At the ED resonance ($\lambda = 572$ nm), the electric field is mainly concentrated close to the sphere with a small presence of the SPP waves propagating on the substrate. However, the propagation of the SPP waves is highly observable at $\lambda = 706$ nm, dominating the scattering behavior of the gold nanosphere. Additionally, the strong interaction between the SPPs' evanescent waves and the gold nanosphere results in cross-coupling between different multipoles, such as MD and EQ, as shown in Figure 2d. It is essential to mention that they possess negative values, yielding destructive interference with the fields scattered from ED resonance coupled to the SPP waves.[89] The SPP waves also produce a strong resonance at $\lambda = 580$ nm in the silicon nanoparticle scattering spectra, as shown in Figure 2e. In this case, the coupling between the SPPs and the particle scattering is so strong that the ED becomes dominant for the whole frequency range, even at the MD resonance ($\lambda = 640$ nm). Although the SPP waves are still coupled to the ED mode at $\lambda = 500$ nm, the contribution of the intrinsic ED resonance of the particle is dominant and more potent compared to the SPP resonance at ($\lambda = 640$ nm), as can be concluded by the near fields presented in the insets of Figure 2e. Once again, the SPP waves couple out of phase to the MD mode, resulting in negative values. Moreover, the MQ resonance weakly couples to the other multipoles or the SPP waves. Finally, for the multilayer particle, the ED is dominant over other contributions, similar to the silicon and gold nanoparticles, as illustrated in Figure 2f. In the ED spectra, there can be seen two different resonances, and the electric field distributions for both are shown in the insets of Figure 2f. For the resonance at $\lambda = 530$ nm, although the propagation of the SPPs is visible, the electric field's major contribution stems from the nanoparticle's ED mode. In contrast, for $\lambda = 629$ nm, surface wave propagation is dominant and controlling the resonance, as shown in Figure 2f. Since both the ED and MD resonances occur at the same wavelength, the cross-coupling between the MD and ED neutralizes the MD resonance of the particle at $\lambda = 530$ nm, resulting in a weak negative contribution.

As shown so far, the plane wave excitation cannot be exploited to selectively excite a multipole resonance mode in a particle, specifically in the presence of a metallic subtrate. Therefore, we propose utilizing the HFOBs containing OAM to excite the desired multipole since their beam profiles can be engineered to overlap with a specific mode of the nanoparticle.

Despite the limited number of multipoles excited in the particles, due to the cross-coupling raised by the substrate, it is essential to perform the calculations with enough multipoles included. Therefore, a careful study of the method's convergence and numerical validation of its accuracy using the Lumerical FDTD package might be found in Section S12, Supporting Information.

3.2. Selective Excitation of Multipoles Using HFOBs Carrying OAM

Having studied the scattering characteristics of the proposed nanoparticles under plane wave illumination in the previous subsection, here, we investigate the possibility of switching between different multipoles in a nanoparticle using structured light. In this sense, we study the selective excitation of multipoles using highly focused vector vortex beams associated with different polarization and OAM states. In addition, we examine the impacts of the presence of the substrate on selective excitation performance.

Prior presenting the selective excitation results, we provide the selection rules based on the polarization and OAM accompanied by the incident light field. The azimuthally and radially polarized beams can be expanded in terms of magnetic and electric multipoles,[17,76] respectively, which stems from their behavioral resemblances. Consequently, the magnetic multipoles of the particle can be excited using the azimuthally polarized beams, also called the magnetic beams in this article, while the radially polarized ones, also called the electric beams, are utilized to excite the electric multipoles.[36] So far, although one can switch from the magnetic multipoles to the electric ones by changing the incident beam's polarization state, another degree of freedom is still needed to change the order of the excited multipole (e.g., dipole or quadrupole). Therefore, we introduce the topological charge associated with the beam as a parameter, which can be tuned to excite a specific magnetic or electric multipole in the particle.[38] In fact, the field profile of the incident beam can be tuned by changing the analogous polarization and OAM states to match the incident field pattern to the desired mode needed to be excited. To further clarify the physics behind the aforementioned rules, the beam profiles for the incident fields and excited modes in three different scenarios are plotted in **Figure 3**. As can be visualized, the incident field pattern behaves similarly to the corresponding desired mode in each case, where for the magnetic (electric) beam with the topological charge of $m = +1$, the incident field correlates with the MD (ED) mode profile, as shown in

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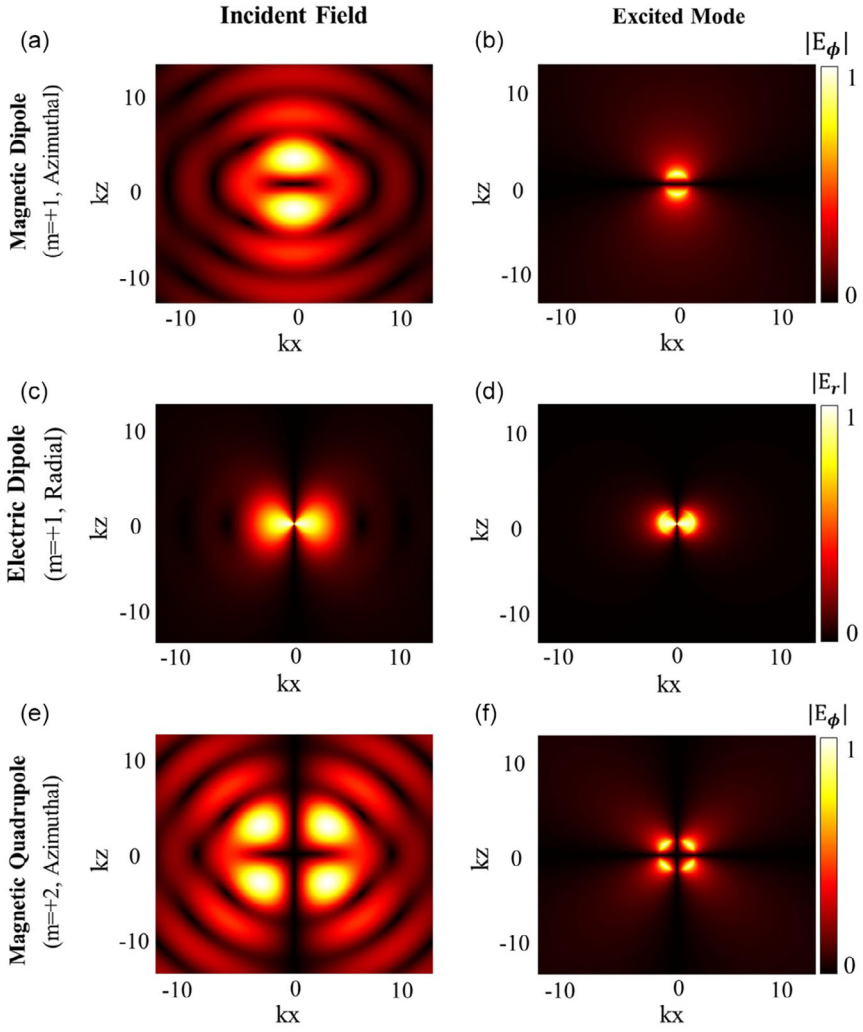


Figure 3. The incident beam and excited multipole field patterns for the magnetic beam illumination with the topological charge of a,b) $m = +1$ and e,f) $m = +2$ and for c,d) the electric beam incidence associated with the topological charge of $m = +1$ for a silicon nanoparticle with a radius of $a = 90$ nm. This figure clarifies the underlying physics behind the selective excitation rules.

Figure 3a–d. However, for the case of the magnetic beam with $m = +2$, the incident field best overlaps with the MQ, as shown in Figure 3e–f. It should also be noted that the Rayleigh length for all the beams in this article is $z_R = \frac{3}{2k}$ in order to ensure the proper interaction with the particles.

3.2.1. Multipole Switching in the Silicon Nanoparticle

After introducing the general rules of selective excitation, we study the scattering properties of the silicon nanoparticle under the illumination of the HFOBs carrying OAM. **Figure 4** shows the total scattering cross section and its multipole decomposition of the silicon particle without (a-c) and with the gold substrate (d–f) for different incident beam configurations: magnetic beam with $m = +1$ (a,d), electric beam with $m = +1$ (b,e), and magnetic beam with $m = +2$ (c,f). In order to quantify the selective excitation performance, we define $\eta_S = |C^D_{sca}/C^T_{sca}|$ as the selective excitation efficiency with C^D_{sca} and C^T_{sca} being the scattering cross section of the desired mode and the total scattering cross section at the resonance wavelength of the desired mode, respectively. As shown, the vortex beam is capable of selectively exciting the MD ($\eta_S = 0.99$) (a), ED ($\eta_S = 0.999$) (b), and MQ ($\eta_S = 0.999$) (c) with negligible disturbance from the undesired modes without the substrate. The only significant interference occurs for the MD excitation that manifests a weak resonance at $\lambda = 445$ nm, corresponding to the MQ resonance mode (see Figure 4a).

Since both MD and MQ modes are excited using the same polarization, the incoming magnetic beam with $m = +1$ used to excite the MD matches imperfectly with the MQ resonance as well, giving rise to the unwanted disruption. The contrary does not occur since the magnetic beam with $m = +2$ used to excite the MQ (Figure 4c) has an inconsiderable overlap with the MD electric field pattern. It should also be noted that the multipole resonance wavelengths do not change compared to the plane wave case since they depend mainly on the geometrical and optical characteristics of the nanoparticle.

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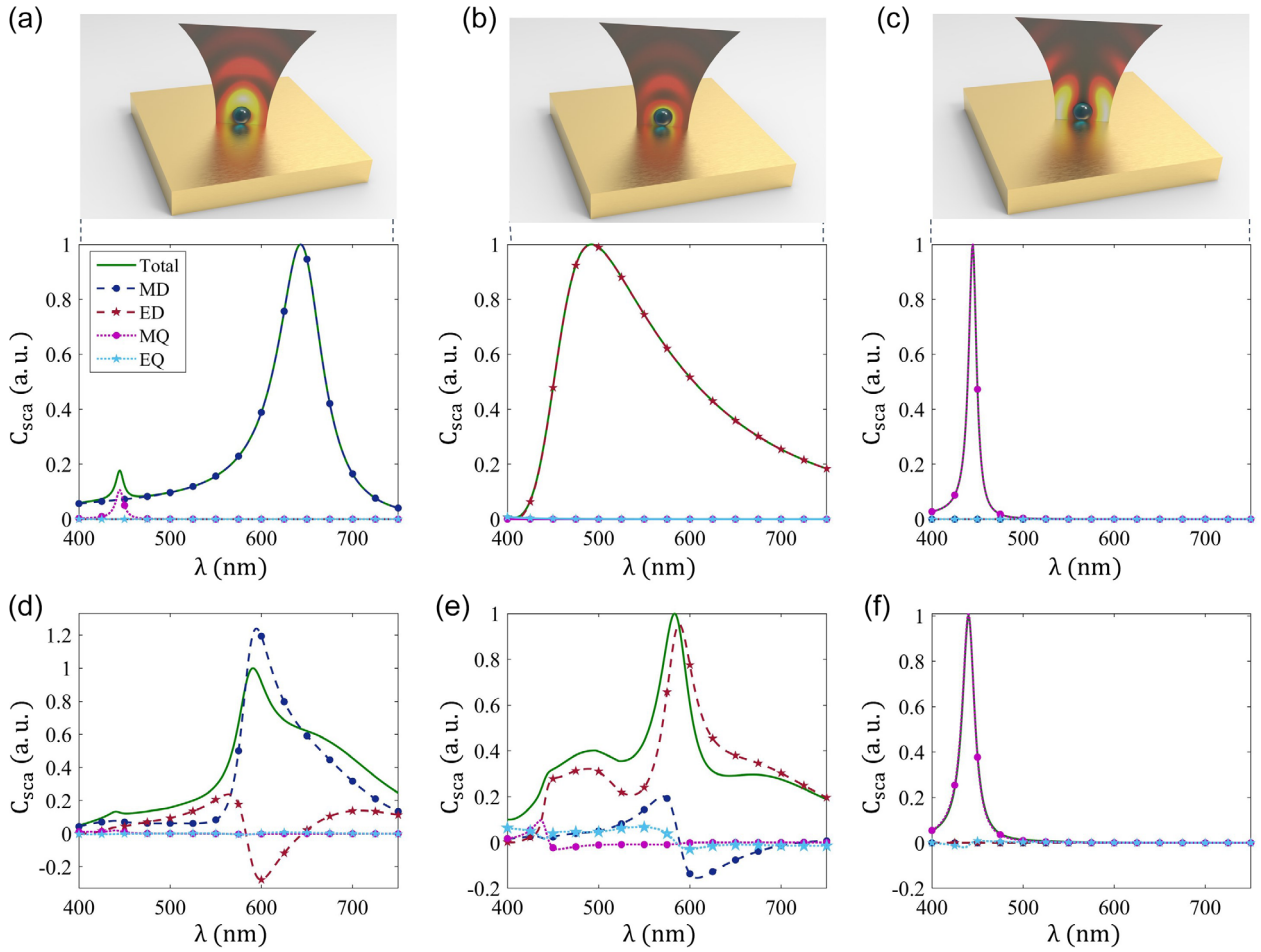


Figure 4. The normalized scattering cross section of the silicon nanoparticle under the illumination of the magnetic beam with the topological charge of a,d) $m = +1$ and c,f) $m = +2$ and b,e) the electric beam associated with the topological charge of $m = +1$ in the a–c) free space and d–f) over a gold substrate. The figure demonstrates the selective excitation performance enabled by the HFOBs carrying OAM in the absence and presence of the metallic substrate.

Finally, we investigate the possibility of selective excitation of multipoles in the particle in the presence of a planar metallic substrate. It is noteworthy that due to the mutual coupling between different multipoles resulting from the presence of the substrate, the perfect selective excitation of multipoles is not expected. However, we demonstrate the possibility of obtaining the desired multipolar behavior in the nanoparticle using the appropriate vortex HFOB.

First, the particle over the substrate is illuminated with the magnetic beam having a topological charge of $m = +1$. Based on the selective excitation rules, one expects the MD resonance to be excited in the particle following its interaction with the incident beam. As shown in Figure 4d, although the resonance is primarily due to the coupling of the evanescent waves to the SPPs ($\lambda = 588$ nm), it reveals its impact majorly in the MD ($\eta_S = 0.97$) mode of the particle. Moreover, while the primary incident field does not excite the ED mode in the particle, the cross-coupling between the MD and the ED as well as the reflection of the incident field will stimulate the ED mode. Besides, the MQ resonance is also observable in the scattering spectra, similar to the particle in free space. Despite different contributions from various multipoles, the MD resonance mode is still dominant in the visible range, particularly at the MD resonance ($\lambda = 640$ nm), where the cross-coupling between the multipoles is negligible, resulting in the presence of the MD mode solely. For the ED excitation over the substrate, as depicted in Figure 4e, the SPPs have the strongest coupling to the ED mode, making it the dominant mode as expected ($\eta_S = 0.93$). Additionally, the MD and MQ resonances are also excited, especially near the SPP resonance ($\lambda = 580$ nm), due to the cross-coupling phenomenon explained in the previous scenario. Finally, the particle over the substrate is illuminated with the magnetic beam with the topological charge of $m = +2$. As can be seen from the scattering spectra shown in Figure 4f, the MQ mode is nearly the only excited multipole in the nanoparticle ($\eta_S = 0.98$) as the primary incident field exclusively stimulates the MQ mode in the particle, which cannot be coupled to the SPPs. Therefore, in this scenario, one will be able to excite only the MQ even in the presence of the substrate.

It is worth mentioning that the acquired characteristics are not achievable using conventional excitation beams, such as a

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plane wave or a highly focused Gaussian beam. The OAM associated with the beam is crucial for engineering a beam profile that is able to overcome the cross-coupling induced by the substrate.

3.2.2. Selective Excitation of Multipoles in a Particle with Tailored Mie Resonances

In this subsection, we study the selective excitation of multipoles in a particle with tailored Mie resonances, which has magnetic and electric dipole resonance modes at the same wavelength. Using a monochromatic plane wave as the incident field, as shown in Figure 2c, it is not viable to distinguish between different resonances. However, using the HFOBs associated with different OAM and polarization states, we selectively excite either the magnetic or electric mode in the multilayer particle. **Figure 5** depicts the total scattering cross section and its decomposition into electromagnetic multipoles of the layered particle without (a-b) and with the gold substrate (c-d) for different OAM configurations: magnetic (a,c), and electric (b,d) beams with $m = +1$. As can be seen from the normalized scattering spectra in Figure 5a, the MD resonance is excited in the particle with zero contribution from the other undesired modes ($\eta_S = 0.999$). Similarly, only the ED ($\eta_S = 0.999$) is excited using the electric beam with the same topological charge, as shown in Figure 5b. As can be seen, for both scenarios, the disturbance from the undesired multipoles is negligible, and only the desired mode is excited without a substrate.

When the substrate is added, the scattered field from the particle couples to the SPPs, resulting in the appearance of the ED even with the magnetic beam illumination, especially close to the SPP resonance at $\lambda = 629$ nm, as demonstrated in Figure 5c. Nonetheless, at the MD resonance ($\lambda = 522$ nm), the MD is dominant ($\eta_S = 0.88$) and much stronger than the ED contribution, denoting the feasibility of selective excitation of the desired multipole even in the presence of the gold substrate. In the case of the electric beam excitation (Figure 5d), the ED is dominant ($\eta_S = 0.91$) as expected due to its strong coupling to the propagating surface waves. Furthermore, the cross-coupling also induces different multipoles (e.g., MD and MQ), which are significantly weaker than the desired resonance needed to be excited (ED).

Finally, we emphasize that in the case of spectral intersection of the resonances, using the structured light to excite them selectively is crucial, which is not attainable using a conventional excitation. Additionally, selective excitation with the HFOBs carrying OAM can be applied to various applications, including increasing

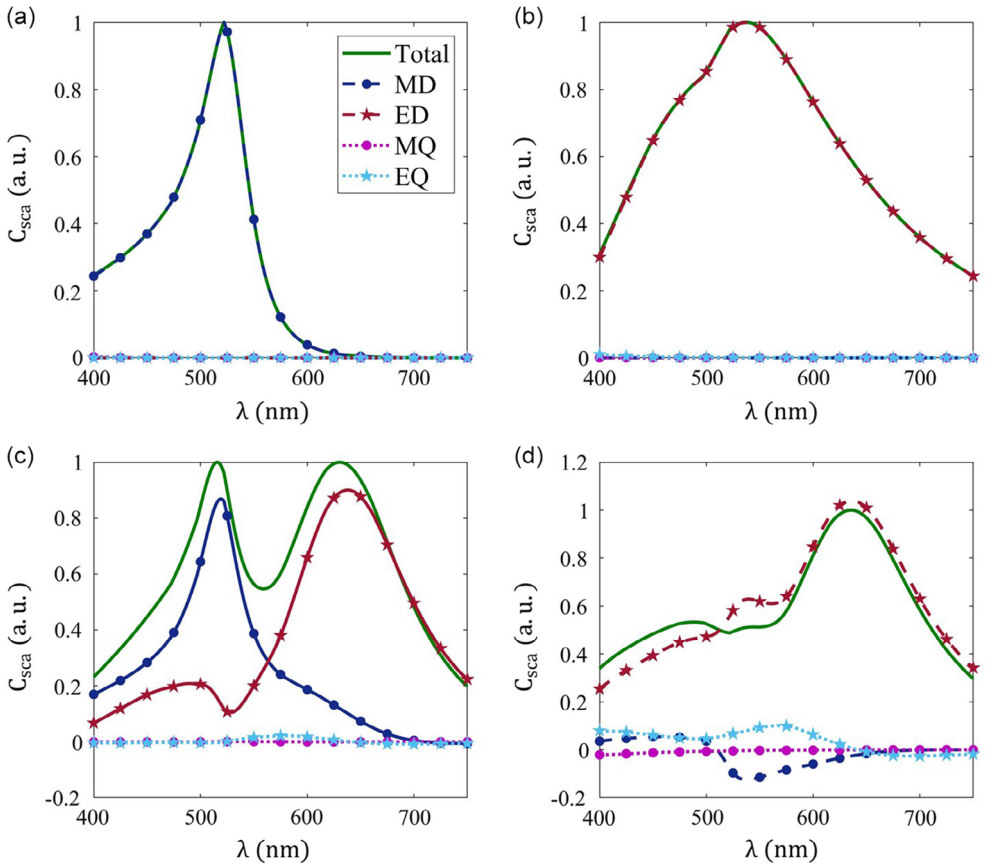


Figure 5. The normalized scattering cross section of the multilayer nanoparticle made of silicon and gold under the illumination of the a,c) magnetic beam with the topological charge of $m = +1$ and the b,d) electric beam associated with the topological charge of $m = +1$ in the a,b) free space and c,d) over a gold substrate. This figure illustrates how the beam profile of HFOBs can be engineered to distinguish between different resonances in particle with tailored Mie resonances.

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the channel capacity in the communications systems.[90,91] In fact, OAM beams associated with different topological charges are mutually orthogonal, making them excellent candidates for increasing the channel capacity of communication systems.[92] Therefore, selective excitation can be utilized to detect the beams with different polarization and OAM states. Moreover, selective excitation can be exploited to control the forces for optical manipulation, which is described in detail in the next section.

4. Optical Manipulation Using Highly Focused OAM Beams

3D trapping of particles located over a metallic substrate is quite challenging due to the image–dipole interaction, giving rise to an attractive force between the particle and the substrate.[93] However, the selective excitation enabled by the HFOBs carrying OAM can open up a new possibility of enhanced optical manipulation, taking advantage of the engineered multipole stimulation. Here, we study the optical forces imposed on the gold and silicon nanoparticles placed over a planar metallic substrate and illuminated by HFOBs associated with different polarization and OAM states. We engineer the incident beam profiles to achieve the desired functionality based on the multipole excited in the particle. The radius for both particles is fixed at $a = 90\,\text{nm}$, and the distance from each center to the substrate is assumed to be h (the axial forces are plotted with respect to the dimensionless variable h/a, which determines how far the particle is placed from the substrate).

First, for the gold nanoparticle, as a highly absorptive object, the axial (F_z) and radial (F_ρ) components of the optical force are depicted in **Figure 6** for different incidence scenarios as a function of wavelength and distance from the substrate. In addition, the black contours determine the change in the optical force's sign, and the particle is slightly shifted from the beam center ($\Delta = 0.2\lambda$) in the transverse plane to make the study of the trapping stability possible (to confirm if the particle is pulled back to the equilibrium position). The gold particle only supports an ED mode in the visible range, splitting into two parts after adding the substrate, where the second resonance is a direct result of coupling the ED mode to the propagating surface waves in the gold–air interface (see Figure 2). Following the illumination of the particle by the magnetic beam associated with the topological charge of $m = +1$ and $m = +2$, the imposed optical forces can be seen in Figure 6a–d. As shown in Figure 6a,c, for both of the cases, the axial force is mainly attractive (negative) near the resonance region ($\lambda = 500$–$750\,\text{nm}$) and in the vicinity of the substrate. It happens on top of the fact that the coupling strength between the SPPs and the particle reaches its maximum value close to the substrate, resulting in a strong attractive force due to the image–dipole interaction between the ED and its image inside the metallic substrate.

By getting far from the substrate, the coupling strength to substrate drops, and the interaction between the primary and secondary (reflected) fields brings out some regions with repulsive force. However, outside the resonance region, where the optical gradient force can compete with the radiation pressure, the axial force is positive (repulsive), making the levitation feasible. In addition, in terms of the radial force, as shown in Figure 6b,d, the force possesses positive (pushing) values with its maximum close to the substrate as the SPPs are propagating in the positive radial direction (diverging waves).[40] Therefore, although the particle can be levitated using the mentioned beams, optical levitation integrated with trapping in three dimensions is not feasible. Besides, when the particle is illuminated by the electric beam accompanied by the topological charge of $m = +1$ and $m = +2$, the ED mode is strongly coupled to the SPPs, resulting in an attractive axial force, which does not allow the trapping as shown in Figure 6e,g. Also, as shown in Figure 6f, h, the radial force expresses pushing features due to the coupling to the SPP waves. Although the particle cannot be levitated from the substrate surface using the current configuration, the highly focused vector vortex beams can offer regions (for instance, $\lambda = 732\,\text{nm}$ and $h = 102\,\text{nm}$ in Figure 6c,d in which it can be trapped stably as will be shown). To overcome the levitation issue, the focus point of the beam can be displaced to a spot below the substrate, which brings out the repulsive axial force in the vicinity of the substrate.[94]

For the silicon nanoparticle, as an example of a high-refractive-index object, the calculated optical forces are shown in **Figure 7** for different scenarios. When the particle is illuminated by the magnetic beam with the topological charge of $m = +1$ and $m = +2$, the MD and MQ are excited, respectively. According to the image theory, the force between a magnet and a metallic substrate is repulsive.[95] Therefore, as shown in Figure 7a,c, the axial force is mainly repulsive close to the magnetic resonances (MD at $\lambda = 445\,\text{nm}$ or MQ at $\lambda = 640\,\text{nm}$). It should also be noted that due to the enhanced coupling in the vicinity of the substrate, the forces around the SPP resonance ($\lambda = 629\,\text{nm}$) decrease by getting far from the substrate. The radial force also shows (negative) pulling behavior around the magnetic resonances and changes its sign in those regions due to the interaction of the reflected scattered field coupled to the SPPs with the particle,[42] as illustrated in Figure 7b,d. Moreover, in the case of the electric beam illumination with the topological charge of $m = +1$ and $m = +2$, as shown in Figure 7e,g, the axial force is primarily attractive due to the coupling of the electric modes to the propagating surface waves. Also, the ED mode has the lowest quality factor among other resonances in the silicon particle, resulting in an attractive axial force in the whole spectrum due to the image–dipole interactions. Furthermore, as can be seen in Figure 7g,h, the radial force is also positive for the mentioned illuminations owing to the scattering of the SPPs, propagating on the surface, from the particle. For the silicon nanoparticle, the HFOBs with OAM can be used to levitate the particle from the substrate and provide stable trapping regions (for instance $\lambda = 584$ nm and $h = 111\,\text{nm}$ in Figure 7a,b in 3D as shown in the following).

It is also worth mentioning that the values of the calculated optical forces provided before are adequately enough for both optical trapping and levitation. More specifically, for the radius considered for the particles under study ($a = 90\,\text{nm}$), the mass is $m_g = 5.9 \times 10^{-17}$ g and $m_{si} = 7.1 \times 10^{-18}$ g for the gold and silicon nanoparticles, respectively. Neglecting the fluctuations, it is only needed to overcome the gravitational force acting on the nanoparticles, which is realizable with a beam of a power on

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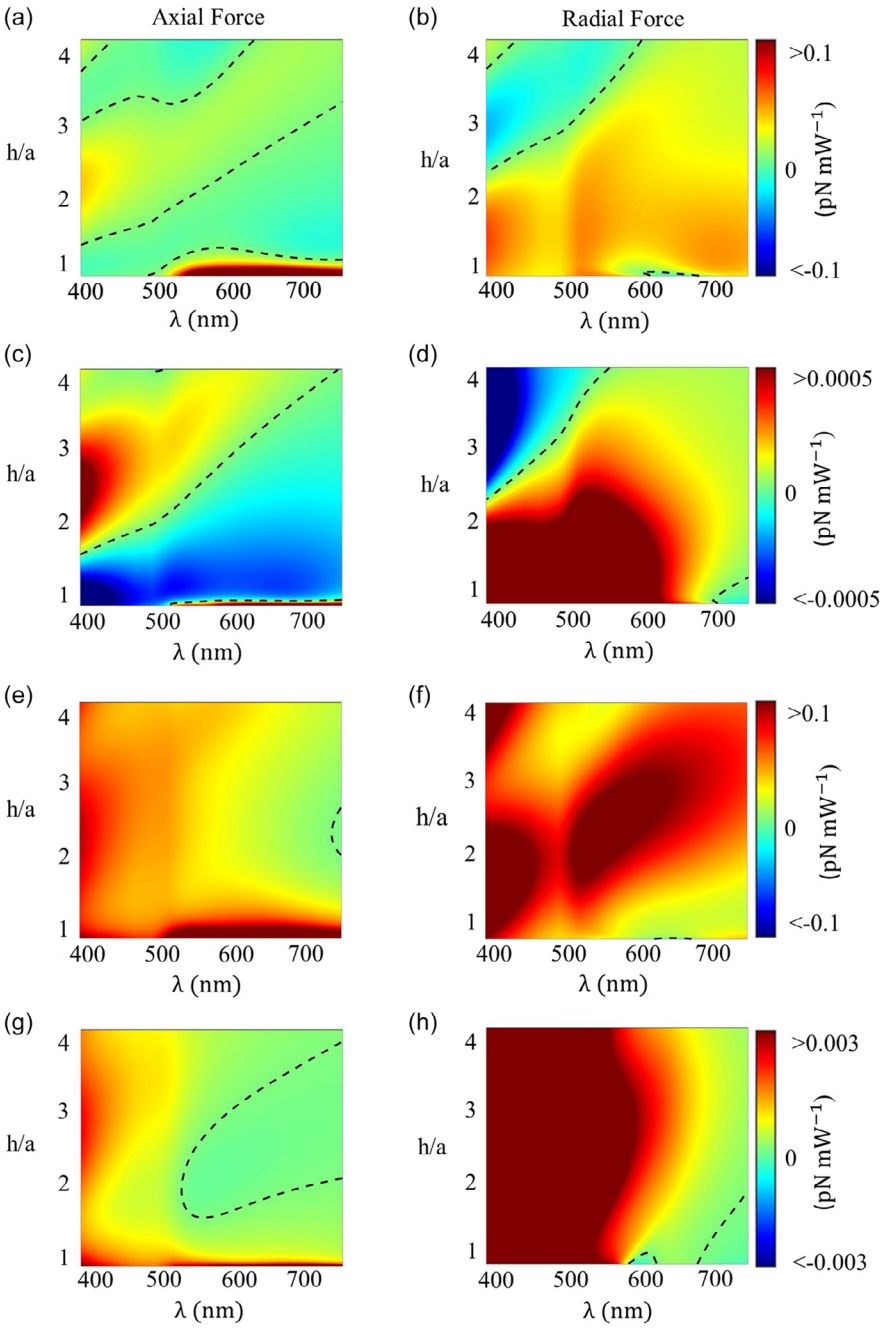


Figure 6. The a,c,e,g) axial and b,d,f,h) radial components of the optical force acting on the gold nanoparticle over the gold substrate for the magnetic illumination with the topological charge of a,b) $m = +1$ and c,d) $m = +2$ and for the electric beam illumination with the topological charge of e,f) $m = +1$ and g,h) $m = +2$. The black plotted contours distinguish between the regions with optical forces with opposite signs. The figure shows the regions where the particle can be trapped either in the axial or radial direction.

the order of mW, which is consistent with the experimental and theoretical works reported before.[40,41,96] It is also worth noting that the trap is resilient to an acceptable extent with respect to thermal fluctuations.

In order to investigate the trapping stability of the particle in the transverse plane, for both the gold and silicon nanoparticles, the radial and azimuthal components of the optical force are shown in **Figure 8**. For the gold nanoparticle, for the case of the magnetic beam illumination with the topological charge of $m = +2$, as shown in Figure 8a, the radial force is negative in a region ($\lambda = 732$ nm and $h = 102$ nm) around the beam center, which can pull the particle to the equilibrium position in the case of initial displacement from the beam center. Figure 8b illustrates the azimuthal component of the optical force where due

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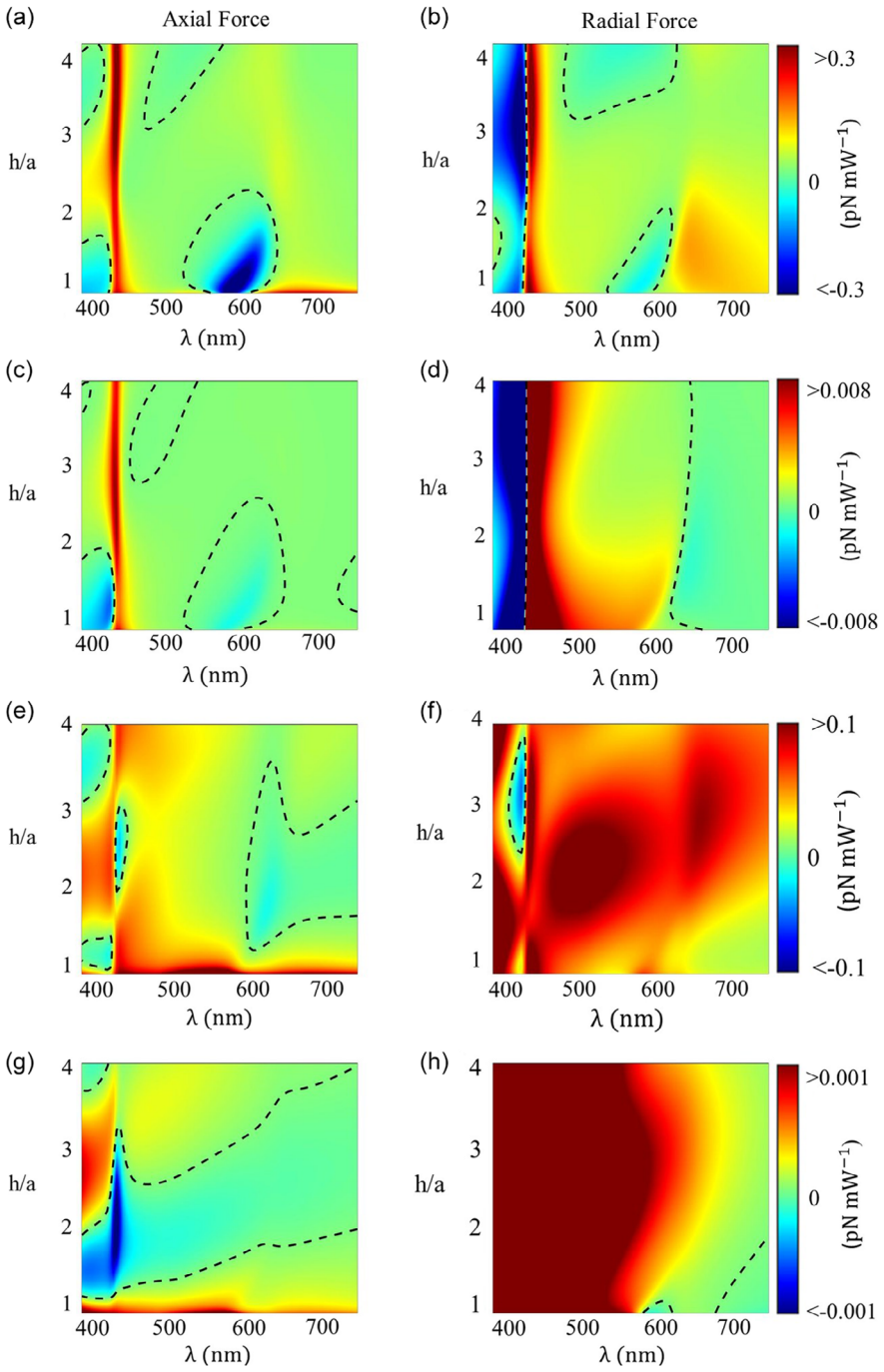


Figure 7. The a,c,e,g) axial and b,d,f,h) radial components of the optical force acting on the silicon nanoparticle over the gold substrate for the magnetic beam illumination with the topological charge of a,b) $m = +1$ and c,d) $m = +2$ and for the electric beam illumination with the topological charge of e,f) $m = +1$ and g,h) $m = +2$. It should be noted that the black contours show the boundaries between the regions where the optical forces change sign. Trapping areas in the normal or transverse directions can be seen in this figure.

to the spiral phase profile of the incident beam, a rotating force in the azimuthal direction is observable. Besides, for the silicon nanoparticle, the radial and azimuthal components of the optical force are shown in Figure 8c,d. As shown in Figure 8c, the radial force provides an area ($\lambda = 584$ nm and $h = 111$ nm), in which the particle will be pulled toward the beam center, satisfying a stable trapping condition requirements. Additionally, as the beam carries OAM, the azimuthal component of the force shows a rotational behavior.[97,98]

Next, we investigate the motion trajectories of the particles to better illustrate the trapping possibility in three dimensions. In the transverse plane, using Newton's second law of motion,

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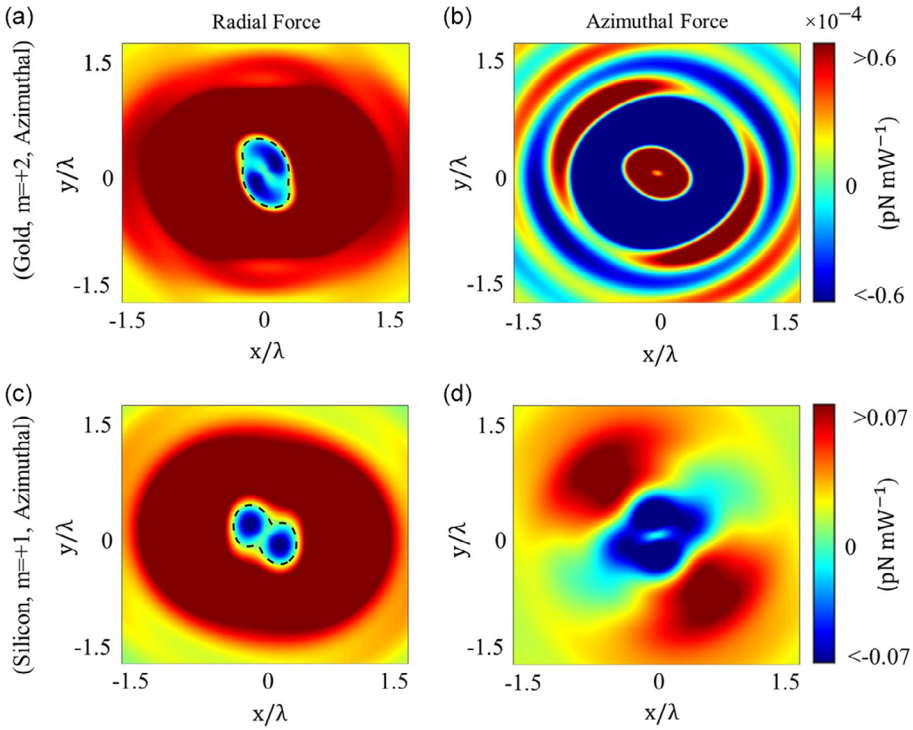


Figure 8. The a,c) radial and b,d) azimuthal components of the optical forces acting on a,b) the gold nanoparticle under the magnetic beam illumination with the topological charge of $m = +2$ at $\lambda = 732$ nm and $h = 102$ nm and for c,d) the silicon nanoparticle under the magnetic beam illumination with the topological charge of $m = +1$ at $\lambda = 584$ nm and $h = 111$ nm. This figure shows the predicted trapping regions for both scenarios.

the dynamics of the particle can be described using the following equation.

$$\mathbf{F}_v - \gamma \mathbf{v} = m\frac{\partial \mathbf{v}}{\partial t} \tag{19}$$

where $\mathbf{F}_v$ is the force induced by the vortex beam, and $\gamma = 6\pi\mu a\,\mathrm{kg\,s^{-1}}$ is the ambient damping constant of air, with $\mu = 1.84 \times 10^{-5}$ being the dynamic viscosity of dry air at room temperature.[42] By integrating Equation (19) over time, the position of the particle can be obtained. For the two mentioned scenarios in Figure 8, the motion trajectories of the particles are depicted in **Figure 9**, statrting from four different initial positions. for four different initial positions (P_1, P_2, P_3, and P_4). For the gold nanoparticle ($m = 19320(4/3)\pi a^3$), as shown in Figure 9a, the strong rotational force generated by the vortex beam can prevent the deviation of the object in the case of surpassing the stable trapping boundaries shown by the black contours. Moreover, for the nanoparticle made of silicon ($m = 2320(4/3)\pi a^3$), as illustrated in Figure 9b, the combination of the radial and azimuthal components of the optical force can stably bring the particle back to the equilibrium. Due to the fact that the total mass of the gold nanoparticle is nearly eight times bigger than the silicon one, the trajectory timeframe of the gold nanoparticle ($t = 2$ ms) is twice of the silicon counterpart($t = 1$ ms). It should be noted that, as the beam with the topological charge of $m = +2$ generates weaker lateral forces with respect to the one with the topological charge of $m = +1$, we have chosen the excitation power of the latter beam multiple times greater than the former to make the trajectory timeframes comparable.

As shown, due to the OAM associated with the incident beam, it can be engineered to selectively excite a specific multipole in the particles placed over a metallic substrate. Consequently, it can offer the possibility of controlling the optical forces imparted to the particles, which can provide stable 3D conditions for both highly absorptive and high-refractive-index objects. It is also worth noting that selective excitation plays a crucial role in controlling the induced optical forces, which is not feasible through a superficial plane wave or Gaussian beam excitation.

5. Conclusion

A physically insightful and numerically efficient analytical platform has been proposed to model the electromagnetic and optomechanical interactions of the highly focused vector vortex beams with nanoparticles over a substrate. The method takes advantage of the GLMT to find the unknown fields and the CSP method to compute the required BSCs of the incident fields. We have applied the introduced technique to depict the viability of engineering the beam profile of an HFOB containing OAM by altering the corresponding polarization and OAM states to excite a specific multipole in a nanoparticle. It has also been demonstrated that although a substrate introduces cross-coupling between different multipoles in nanoparticles, the selective excitation would maintain functionality after adding the substrate to the problem. By resorting to a particle with tailored Mie

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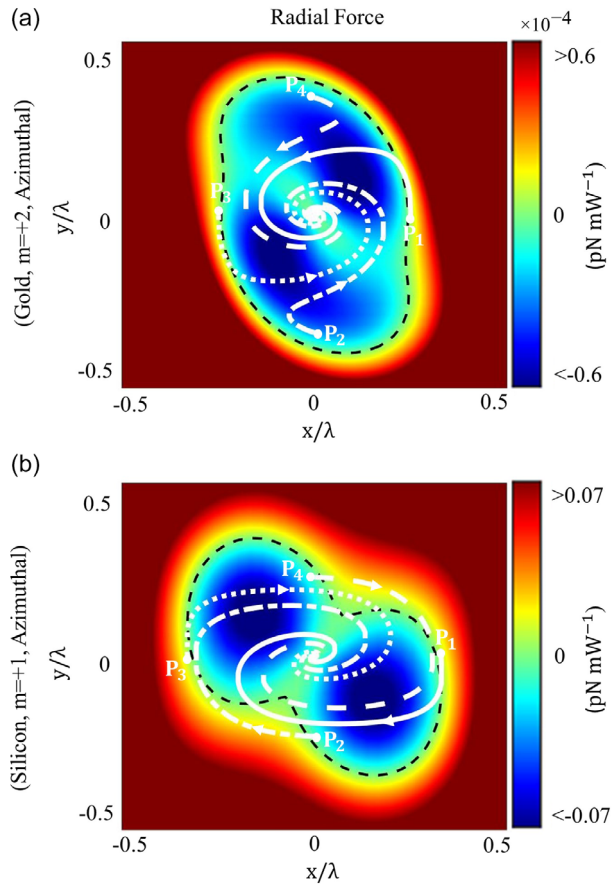


Figure 9. The motion trajectories of a) the gold nanoparticle under the magnetic beam illumination with the topological charge of $m = +2$ and for b) the silicon nanoparticle under the magnetic illumination with the topological charge of $m = +1$. It should be noted that different line styles represent the trajectory of the particle from different initial positions.

resonances, we have demonstrated that the engineered beam profiles of the highly focused vector vortex beams can be exploited to distinguish between the resonances in case they are placed at the same wavelength. Due to the strong coupling of the ED to the SPPs propagating on the surface of the metallic substrate, ED will be the dominant excited mode in a particle in the whole visible range following the plane wave illumination, resulting in an attractive force on top of the image–dipole interaction. It has been illustrated that the parameters of the incident HFOBs carrying OAM can be appropriately chosen to overcome the mentioned issue. We have engineered the incident fields to excite the magnetic modes in particles over the metallic substrate to ensure 3D optical trapping. In conclusion, the performed motion trajectory analyses declared the prosperous advantages of using a highly focused vector vortex beam in the optical manipulation and trapping of nanoparticles as a glance into the potential applications.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Keywords

Mie scattering, optical manipulation, orbital angular momentum, selective excitation of multipoles

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